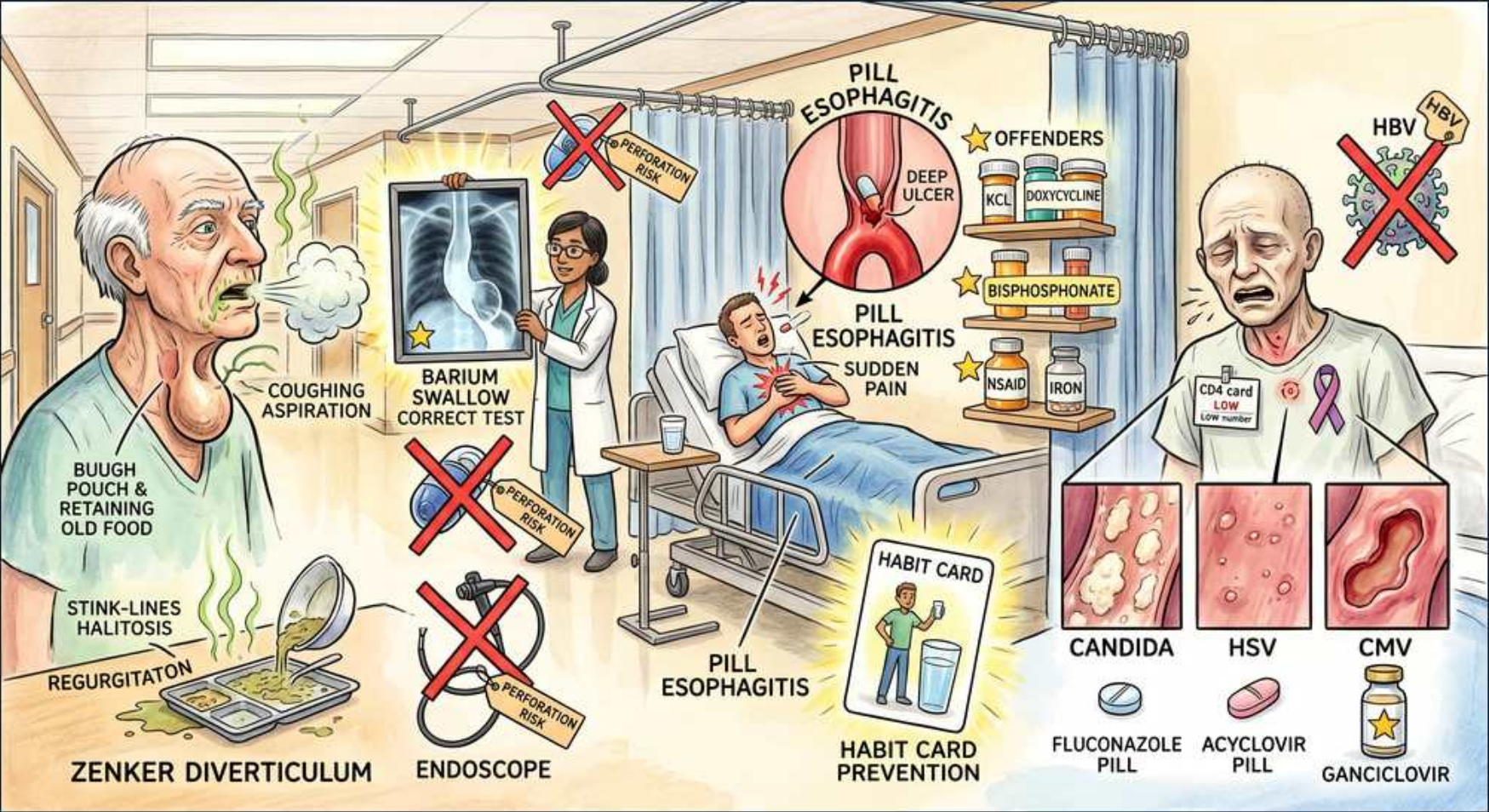


Esophageal — Zenker, Pill & Infectious



1. Zenker diverticulum (pouch, old food)

WHAT YOU SEE

Elderly man with a 'BULGING POUCH & RETAINING OLD FOOD' label, 'ZENKER DIVERTICULUM'.

WHAT IT ENCODES

Zenker diverticulum = a pulsion (false) outpouching through Killian's triangle above a dysfunctional cricopharyngeus muscle. Older patients. Food collects in the pouch and is regurgitated later. Caused by cricopharyngeal incoordination/high pressure.

2. Halitosis / regurgitation

WHAT YOU SEE

Stink-lines and labels 'HALITOSIS' and 'REGURGITATION' by the man.

WHAT IT ENCODES

Classic Zenker tetrad: halitosis (rotting retained food), regurgitation of undigested old food, dysphagia, and a sensation of neck fullness/gurgling. Stagnant food explains the bad breath, a key exam clue.

3. Aspiration / coughing

WHAT YOU SEE

Cough cloud labeled 'COUGHING / ASPIRATION' near the patient.

WHAT IT ENCODES

Regurgitated retained food can be aspirated, causing recurrent aspiration pneumonia and nocturnal cough, especially when supine. Aspiration is the major morbidity of an untreated Zenker.

4. Barium swallow = correct test (gold star)

WHAT YOU SEE

X-ray film with doctor, banner 'BARIUM SWALLOW CORRECT TEST', gold star.

WHAT IT ENCODES

Diagnose Zenker with a BARIUM esophagram (video-fluoroscopic swallow), which safely outlines the posterior pouch. It is the correct first test. Treatment of symptomatic Zenker is cricopharyngeal myotomy +/- diverticulectomy (open or endoscopic).

5. Endoscopy / capsule first (RED X)

WHAT YOU SEE

Endoscope and capsule icons each crossed out with 'PERFORATION RISK' red X labels.

WHAT IT ENCODES

Blind passage of an NG tube, endoscope, or video capsule FIRST in suspected Zenker risks intubating and PERFORATING the pouch. Get the barium study first; if endoscopy is later needed it is done carefully/under direct vision.

BOARD TRAP

'endoscopy or video capsule first in suspected Zenker' -> wrong; blind passage can perforate the pouch; barium swallow first.

6. Pill esophagitis = deep focal ulcer

WHAT YOU SEE

Cross-section showing a 'DEEP ULCER' with banner 'PILL ESOPHAGITIS'.

WHAT IT ENCODES

Pill (medication-induced) esophagitis: a pill lodges and erodes the mucosa, causing a deep, focal/localized ulcer, typically at the mid-esophagus (aortic arch/LA impression). Presents with sudden retrosternal odynophagia and chest pain.

7. Sudden odynophagia

WHAT YOU SEE

Patient in bed clutching chest, 'SUDDEN PAIN' burst, 'PILL ESOPHAGITIS'.

WHAT IT ENCODES

Hallmark history: abrupt onset of severe odynophagia/retrosternal pain shortly after taking a pill, often with too little water or while supine/at bedtime. Usually self-limited once the offending habit is corrected.

8. Offenders: KCl, doxy, bisphosphonate, NSAID, iron

WHAT YOU SEE

Pill bottles labeled 'KCL, DOXYCYCLINE, BISPHOSPHONATE, NSAID, IRON' with star, banner 'OFFENDERS'.

WHAT IT ENCODES

Common culprits: potassium chloride, tetracyclines/doxycycline, bisphosphonates (alendronate), NSAIDs, iron, ascorbic acid, quinidine. Each is caustic when lodged. Recognize the drug list on a stem with sudden odynophagia.

9. Habit card: pills upright with water (gold star)

WHAT YOU SEE

Figure on a card 'HABIT CARD PREVENTION', gold-star, person standing upright.

WHAT IT ENCODES

Prevention = take pills with a full glass of water and remain UPRIGHT (sitting/standing) for >=30 min, avoid taking at bedtime. This is the management/prevention answer; the esophagitis itself heals with supportive care once the habit changes.

10. Candida = white plaques (fluconazole)

WHAT YOU SEE

Endoscopy panel 'CANDIDA' with whitish plaques and a 'FLUCONAZOLE PILL' below.

WHAT IT ENCODES

Candida esophagitis: linear adherent white plaques; common with immunosuppression/HIV (and inhaled steroids). Often with oral thrush. Treat with systemic fluconazole. Most common infectious esophagitis.

11. HSV = small punched-out ulcers (acyclovir)

WHAT YOU SEE

Endoscopy panel 'HSV' with small punched-out ulcers and 'ACYCLOVIR PILL'.

WHAT IT ENCODES

HSV esophagitis: multiple small, shallow, 'punched-out' (volcano) ulcers; biopsy from ulcer EDGE shows Cowdry A inclusions/multinucleated giant cells. Treat with acyclovir. Seen in immunocompromised (occasionally immunocompetent).

12. CMV = large/deep linear ulcers (ganciclovir, gold star)

WHAT YOU SEE

Endoscopy panel 'CMV' with a large deep linear ulcer and gold-star 'GANCICLOVIR' vial.

WHAT IT ENCODES

CMV esophagitis: large, deep, linear/serpiginous ulcers in advanced immunosuppression (HIV with CD4 <50, transplant); biopsy from ulcer BASE shows intranuclear 'owl-eye' inclusions. Treat with ganciclovir (or foscarnet/valganciclovir).

13. Low CD4 context

WHAT YOU SEE

Patient holding a 'CD4 CARD LOW' with an immunosuppression ribbon.

WHAT IT ENCODES

Infectious esophagitis flags immunosuppression: candida and HSV at moderate immunodeficiency, CMV at very low CD4 (<50). Odynophagia in an HIV patient -> think infectious esophagitis; CD4 count helps predict the organism.

14. HBV as a cause (RED X)

WHAT YOU SEE

Virus icon labeled 'HBV' with a red X.

WHAT IT ENCODES

Hepatitis B virus does NOT cause esophagitis; the viral causes of infectious esophagitis are HSV, CMV and VZV (plus fungal Candida). HBV on an esophagitis stem is a distractor.

BOARD TRAP

'HBV as a cause of esophagitis' -> wrong; viral esophagitis is HSV/CMV/VZV (and Candida); HBV is not a cause.